



200 series

SH-ISENH4200

Ammonia digital probe

User manual

Reading instructions

Information for users

Thank you very much for choosing to use our SH-ISENH4200 Ammonia online digital probe (hereinafter referred to as: NH4200 probe). Before using this product, please read this user manual carefully. This manual covers important information and data for the use of the product, which must be strictly observed by the user to ensure the normal operation of the NH4200 probe. The information will help you to use the product correctly and to obtain accurate analysis results.

General description

This manual provides a detailed description of the installation, commissioning, operation and maintenance of the NH4200 probe, as well as describing the measurement principle, instrument composition and performance characteristics of the NH4200 probe.

This manual instructs the user on the proper installation and operation of the NH4200 probe, as well as preventive maintenance work on the NH4200 probe to guarantee the continuous and reliable operation of this sensor.

The products described in this manual have been rigorously inspected before leaving the factory to ensure top quality. In order to ensure safe, high-quality operation and to obtain correct analysis results, the user must strictly follow the instructions described in this manual.

This manual provides all the information required for the proper use of the NH4200 probe. It provides an accurate reference for use by technicians with specialized training or knowledge of the instrument's operational controls. Before operating the instrument, make sure that the safety and warning messages mentioned in this manual are correctly understood and applied to the actual operation.

Delivery and transport

Shipment requirements are in accordance with the terms of the ordering contract.

Please read the appropriate information on the packaging material carefully when opening the box to ensure that the opened goods are complete and undamaged. Please try to retain the outer packaging of the product in case you need to return the meter or parts.

Warranty and maintenance

1) SYSTEALOG warrants all Products manufactured to be free from defects in material and workmanship under normal use and service. SYSTEALOG's obligation under any warranty is limited to and Buyer's exclusive remedy hereunder shall be, repair or replacement, Ex Works SYSTEALOG's factory, of any defective parts, when returned to SYSTEALOG by Buyer, transportation prepaid, which SYSTEALOG's examination discloses to have been factory defective. The time limit of this warranty is 12 (twelve) months from date of shipment of new Products unless otherwise specified by SYSTEALOG. SYSTEALOG shall not be liable for consequential damages, in any event. SYSTEALOG will decide at its only discretion whether to repair or replace a defective part.

For wearing parts and/or in contact with aggressive substances Sys tea reserves to reduce the warranty period accordingly to the specific application.

2) Repaired Products, all repair work done by SYSTEALOG is warranted to be free from defects in material or workmanship under normal use and service. SYSTEALOG's obligation under this warranty is limited to, and the exclusive remedy hereunder shall be, repair or replacement, Ex Works SYSTEALOG's factory, of any defective parts installed by SYSTEALOG, when returned to SYSTEALOG, transportation prepaid, which SYSTEALOG's examination discloses to have been factory defective. The time limit of this warranty is three (3) months from date of shipment. SYSTEALOG shall not be liable for consequential damages, in any event. SYSTEALOG will decide at its only discretion whether to repair or replace a defective part.

3) THESE WARRANTIES ARE IN LIEU OF ANY WARRANTY OF MERCHANTABILITY, OR FITNESS, OR ANY OTHER WARRANTY, EXPRESSED OR IMPLIED.

4) Expendable Items: this Warranty does not cover expendable or consumable items such as lamps, fuses, parts in contact with fluids, etc.

5) SYSTEALOG has reasonable period of time for replacements/repairs, and cannot liable for any compensation may arise in relation to the Product

6) SYSTEALOG only obligation in case of defects, lack of quality or non-conformity of the Products will be repairing or replacing the defective parts. It is agreed that the above-mentioned warranty (i.e.: the obligation to repair or replace the defective parts) is in lieu of any other legal guarantee or liability with the exclusion of any other SYSTEALOG liability (whether contractual or non-contractual) which may anyhow arise out of or in relation with the Products supplied (e.g. compensation of damages, loss of profit, recall campaigns, etc.).

8) The warranty on goods supplied by SYSTEALOG is void after 12 months for new Products, if not otherwise specified, 3 months for repairs from delivery date, and in case of one of the following conditions:

- a) Due to improper use (water ingress, corrosion, fire, strong electricity strung in, etc.);
- b) Damage caused by force majeure (earthquakes, lightning strikes, floods, etc.);
- c) Unauthorized alterations within the product without permission;
- d) Failure to use the product in accordance with the user's manual and training provisions, causing damage to the product.
- e) On products not manufactured by SYSTEALOG but sold by SYSTEALOG as results of a distribution contract or as occasional resale, apply the warranty conditions of the manufacturer of the product.
- f) Any warranty request, within specified limits, must be notified, the SYSTEALOG using the form "Warranty Claim" by fax, mail or registered mail, within seven (7) days from the event of the defect. Failing such notification, the Purchaser's right to claim the defects will be forfeited.

Summary

READING INSTRUCTIONS.....	2
INFORMATION FOR USERS	2
GENERAL DESCRIPTION	2
DELIVERY AND TRANSPORT	2
WARRANTY AND MAINTENANCE	2
1 INSTRUMENT OVERVIEW	6
1.1 INSTRUMENT INTRODUCTION	6
1.2 MEASUREMENT PRINCIPLE.....	6
1.3 PRODUCT FEATURES	7
1.4 MAIN USES AND SCOPE OF APPLICATION	7
2 TECHNICAL SPECIFICATION.....	8
3 INSTALLATION AND OPERATION.....	9
3.1 INSTALLATION.....	9
3.2 WIRING INSTRUCTIONS	10
3.3 COMMUNICATION PROTOCOL	11
3.4 MODBUS-RTU PROTOCOL EXAMPLE (ADDRESS DEFAULT 0x32)	16
3.4.1 <i>Reading the measured value</i>	16
3.4.2 <i>pH1 calibration</i>	16
3.4.3 <i>pH2 calibration</i>	16
3.4.4 <i>pH3 calibration</i>	17
3.4.5 <i>Ammonia 1 calibration</i>	17
3.4.6 <i>Ammonia 2 calibration</i>	17
3.4.7 <i>Temperature calibration</i>	18
4 USE AND MAINTENANCE	19
4.1 PROBE ACTIVATION.....	19



4.2	CALIBRATION	19
4.2.1	<i>pH calibration</i>	19
4.2.2	<i>Ammonia calibration</i>	20
4.3	CONDITIONING.....	20
4.4	MAINTENANCE.....	20
4.5	TROUBLESHOOTING	21
6	ADDITIONAL INFORMATION	22
6.1	UNPACKING AND INSPECTION	22
6.2	SAFEKEEPING	22
6.3	PRECAUTIONS TO USE	22
6.4	AFTER SALE SERVICE	22

1 Instrument overview

1.1 Instrument Introduction

The Ammonia probe is capable of real-time monitoring of three parameters: temperature, pH, and ammonia concentration in water. The probe consists of sensors (including temperature sensors, pH electrodes and ammonium ion electrodes), circuits (internal), and structural components, the appearance of which is shown in Figure 1.1.

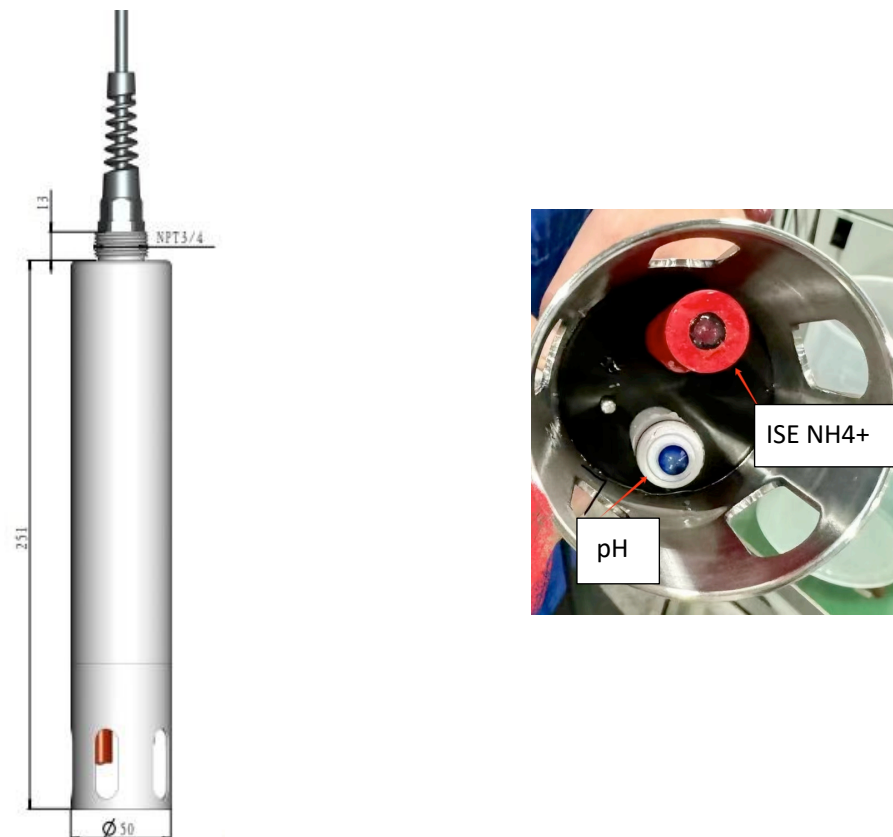


Figure 1.1 Appearance of the Ion Selective Electrode Ammonia probe

1.2 Measurement principle

This product adopts ion-selective electrode method to measure ammonia nitrogen (NH_3) in water body, including temperature, pH and ammonia nitrogen. Temperature is measured by NTC temperature sensor, the detected voltage of different temperatures is processed by the circuit and converted to the corresponding temperature value by the instrument. pH is measured by glass electrode, the potential of the electrode is collected and converted to pH value by Nernst's equation. Ammonia measurement is calculated by Ammonium (NH_4^+) concentration measured by an Ion Selective Electrode and the other two measurements.

The ammonium ion selective electrode has good selectivity for NH_4^+ in water, and the corresponding

electrode potential is obtained through electrochemical reaction. The potential is converted to the concentration of NH_4^+ by the Nernst equation.

Temperature compensation of pH value based on measured temperature value; temperature and pH compensation of ammonia nitrogen concentration by temperature and pH value to obtain more accurate ammonia nitrogen concentration.

The SH-ISENH4200 probe doesn't include Potassium ion (K^+) internal compensation; its interference ratio versus the measured ammonium ion (NH_4^+) concentration is 10:1.

At this ratio, there will be approximately a 10% error in the final measurement result. Depending on the concentrations to be measured, the SH-ISENH4200 probe must be calibrated on a specific decade approximately around the expected values, for example: 2-20 mg/L NH_3 for samples around 10-15 mg/L or 50-500 mg/L NH_3 for samples around 200-300 mg/L; along the selected calibration decade, the measurement concentration will be as accurate as in the data-sheet; in any case, the probe will provide measurement data in the whole measuring range as declared in data-sheet.

Accordingly, the operative detection limit would be strictly connected to the selected calibration decade.

1.3 Product features

- pH and temperature-compensated electrodes for a wide range of sites
- Each electrode can be replaced independently, simple operation and low maintenance
- Fast response time, real-time detection of ammonia nitrogen in water bodies
- Corrosion-resistant housing, waterproof grade IP68, can work underwater for a long time
- RS485 communication interface, standard Modbus protocol for easy integration.

1.4 Main uses and scope of application

Environmental surface water, municipal and industrial wastewater, industrial process water.

2 Technical specification

Working Principle	Ion Selective Electrode method		
Measuring parameter	Ammonia	pH	Temp
Meas. range	(0.05~1000) mg/L	0~14	(0~50) °C
Repeatability	≤1 %	<0.1 pH	0.1 °C
Accuracy	±5% or ±0.1mg/L	<0.1 pH	0.1 °C
pH comp. range	4~10		
Zero drift (24h)	±1% F.S.		
Range drift (24h)	±3% F.S.		
Precision	Ammonia: 1-1000mg/L: ±5% 0.1-1mg/L: ±0.1mg/L		
Detection limit	0.05 mg/L		
Resolution	0.01 mg/L		
protection class	IP68		
Comm interface	RS485, standard Modbus protocol Baud rate 9600		
Dimensions	(φ50*251) mm, cable 10 m (customizable)		
Materials	316 Stainless-steel/titanium alloy available		
Weight	2 kg		
Operating voltage	9~36 Vdc		
Power absorption	0.5 W		
Calibration period	1~3 months (depending on site water quality environment)		
Operating temp.	(0~50) °C		

Table 2.1

3 Installation and operation

3.1 Installation

The installation of the ion-selective electrode ammonia probe (subsequently referred to as the probe) has a greater impact on its measurement, and the environment in which it is installed and used needs to meet the following requirements:

- Before installation and use, turn the probe part down vertically and shake it gently to make the electrolyte inside the ammonium ion electrode fill the head of the membrane; after that, the sensor part should always be kept down vertically, and the angle of inclination should not be greater than 45° to prevent the ammonium ion electrode from generating air bubbles at the head of the membrane and causing data anomalies.



Figure 3.1 Flicking the sensor site

- When installing the probe, make sure that the surrounding area is not damaged by external collisions, cuts and pressure.
- The minimum water level should be submerged in the probe after installation to prevent the electrode from contacting the liquid to be measured when the water level is too low. In addition, it is necessary to ensure that the electrode is in a liquid environment for a long time after installation to prevent the electrode from being damaged by a long period of time without water.
- The flow velocity of the water during measurement is not more than 3.0m/s, without abrasive media.
- For submerged installation, the probe must be mounted on the mounting bracket and must not be suspended in water by the sensor cable.

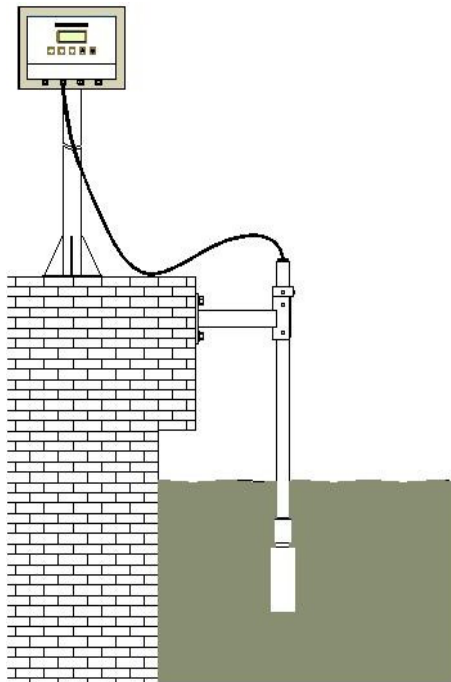


Figure 3.2 Schematic diagram of submerged installation

3.2 Wiring instructions

Sensor type	Wire color	Meaning	Connection type
ISENH4200 ammonia digital probe	red	Power positive (+)	24 Vdc power supply
	black	Power negative (-)	24 Vdc power supply
	blue	485A	RS-485 communication
	yellow-brown	485B	RS-485 communication
	violet	PE	grounding

Table 3.1 Wiring description

Sensor wiring, strip the end of the cable to reveal 5 wires, please check the table 3.1 were connected to the corresponding port, any short circuit may cause the instrument burned, data loss or program damage! Confirmation of the correct wiring can be connected to the power to use.

WARNING: The PE probe wire must be connected to the PE terminal of the controller.

CAUTION: Do not connect the PE probe wire to the main power terminals (L and N), otherwise serious damage to the probe or controller may occur. Carefully check the probe wire connection before turning on the controller

3.3 Communication protocol

baud	9600
data bit	8
stop bit	1
parity check	None
slave address	0x32

Table 3.2 Serial Port default Parameter Configuration

Modbus-RTU mode, data transfer size end description:

data type	Big and small end model
char (computing)	the main points
uint16	the main points
uint32	Word-ordered little end, byte-ordered big end within word (CDAB)
Float (encoding format IEEE754)	Word-ordered little end, byte-ordered big end within word (CDAB)

Fault code	Reason for failure
1ul<<1	Temperature difference greater than 5°C during pH calibration
1ul<<2	Abnormal slope for pH calibration
1ul<<4	Ammonia nitrogen calibration with a temperature difference of more than 5°C
1ul<<5	Abnormal slope during ammonia calibration
1ul<<6	Ammonia nitrogen concentration greater than set concentration
1ul<<8	Temperature less than 0 or more than 50
1ul<<9	Failure to calibrate actual water samples

Table 3.3 Fault table

Note:

<< is a left-shift operator that first converts the alarm code to binary, with each bit representing an alarm state

For example, 44 is converted to 00000000 00000000 00000000 00101100, which corresponds to fault codes 1ul<<2,1ul<<3,1ul<<5

Functionality	Typology	Meas. unit	Range of values	Byte count	Input register address	Note
Device SN number	ASCII	--	--	12	0x7530	
software version number	ASCII	--		32	0x7536	
Probe Type	uint16		--	2	0x7546	Fixed at 1 for Ammonia
Probe Status	uint16	--	0-5	2	0x7566	0: Measurement 1: pH1 is being calibrated 2: pH2 is being calibrated 3: pH3 is being calibrated 4: Ammonia 1 is being calibrated 5: Ammonia 2 is being calibrated 6: Actual water sample calibration
error code	uint32	--	--	4	0x7567	Normal state is 0, see fault table for details
NH3 concentration	float	mg/L	--	4	0x7569	
temp	float	°C	0-50	4	0x756b	
pH	float	pH	--	4	0x757d	
pH voltage	float	V	--	4	0x757f	
NH3 voltage	float	V	--	4	0x7571	
Temperature voltage	float	V	--	4	0x7573	
pH calibration voltage value 1	float	mV		4	0x7575	
Temperature at pH calibration 1	float	°C	0-50	4	0x7577	
pH calibration voltage value 2	float	mV	--	4	0x7579	
Temperature at pH calibration 2	float	°C	0-50	4	0x757b	
pH calibration voltage value 3	float	mV	--	4	0x757d	

Functionality	Typology	Meas. unit	Range of values	Byte count	Input register address	Note
Temperature at pH calibration 3	float	°C	0-50	4	0x757f	
NH3 calibration voltage Value 1	float	mV	--	4	0x7581	
Temperature during NH3 calibration 1	float	°C	0-50	4	0x7583	
NH3 calibration voltage Value 2	float	mV	--	4	0x7585	
Temperature during NH3 calibration 2	float	°C	0-50	4	0x7587	
Ammonia nitrogen actual water sample calibration voltage value	float	mV	--	4	0x7589	
Temperature at calibration of actual water samples for ammonia nitrogen	float	°C	0-50	4	0x758b	

Table 3.4 Input register protocols

R/W properties: phrase marked by pause

Functionality	Read and Write Properties	Typology	Meas. unit	Range of values	Occupied byte	Holding Register Address	Note
Device address	Read/Write	uint16	--	0 to 247	2	0x9c40	The default value is 0x32, the device is reset after writing, so it must be written to a single register.
Baud	Read/Write	uint32	--	2400. 4800. 9600. 14400. 19200. 38400. 57600. 115200.	4	0x9c41	Default value 9600
Restore Factory Settings	Write	uint16	--	0-1	2	0x9c51	Automatic reset after writing
Restart SW	Write	uint16	--	0-1	2	0x9c53	Automatic reset after writing
pH calibration time1	Read/Write	uint32	s	--	4	0x9c62	UNIX timestamp (calibration starts after writing)
pH calibration value1	Read/Write	float	--	0-14	4	0x9c64	
pH calibration time2	Read/Write	uint32	s	--	4	0x9c66	UNIX timestamp (calibration starts after writing)
pH calibration value2	Read/Write	float	--	0-14	4	0x9c68	
pH calibration time 3	Read/Write	uint32	s	--	4	0x9c6a	UNIX timestamp (calibration starts after writing)
pH calibration value 3	Read/Write	float	--	0-14	4	0x9c6c	
NH3 calibration time1	Read/Write	uint32	s	--	4	0x9c72	UNIX timestamp (calibration starts after writing)
NH3 calibration value1	Read/Write	float	--	0-1000	4	0x9c74	
NH3 calibration time2	Read/Write	uint32	s	--	4	0x9c76	UNIX timestamp

							(calibration starts after writing)
NH3 calibration value2	Read/Write	float	--	0-1000	4	0x9c78	
Temperature calibration value	Read/Write	float	°C	--	4	0x9c7a	
Ammonia exceeds upper limit	Read/Write	float	mg/L	--	4	0x9c7c	
Salinity compensation factor	Read/Write	float	--	--	4	0x9c80	
Electrical conductance	Read/Write	float	--	--	4	0x9c82	
Ammonia nitrogen actual water sample calibration time	Read/Write	uint32	s	--	4	0x9c84	UNIX timestamp (calibration starts after writing)
Ammonia nitrogen actual water sample calibration value	Read/Write	float	--	0-1000	4	0x9c86	
Actual water sample switch	Read/Write	uint32	--	0-1	4	0x9c88	0: Off 1: Open

Table.3.5 Holding registers

3.4 MODBUS-RTU protocol example (address default 0x32)

3.4.1 Reading the measured value

Send: 32 04 75 69 00 06 bf db

0x32:Device address

0x04:Read input register function code

0x7569:Input register start address

0x06:read 6 registers

0xbfdb:Check digits

Received: 32 04 0c dd 0d 42 f6 be d2 41 da 33 33 41 03 df f4

0x32:Device address

0x04:Read input register function code

0x0c:12 bytes

0xdd0d42f6:Ammonia 0x42f6dd0d (123.43)

0xbcd241da:temperature 0x41dabed2 (27.34)

0x33334103: pH 0x41033333 (8.2)

0xbfdb:Check digits

3.4.2 pH1 calibration

Send: 32 10 9c 62 00 04 08 65 35 64 f9 51 ec 40 80 ba f8

0x32:Device address

0x10:Write Multiple Register Function Code

0x9c62:Write register start address

0x04:Write to 4 registers

0x08:Write 8 bytes

0x653564f9:timestamp 0x64f96535(1694065973)2023-09-07 13:52:53

0x51ec4080: calibration value 0x408051ec (4.01)

0x07bd:Check Digit

Received: 32 10 9c 62 00 04 4b 87

0x32:Device address

0x10:Write Multiple Register Function Code

0x9c62:Write register start address

0x04:Write to 4 registers

0x4b87:Check Digit

3.4.3 pH2 calibration

Send: 32 10 9c 66 00 04 08 69 50 64 f9 85 1f 40 db f6 5c

0x32:Device address

0x10:Write Multiple Register Function Code

0x9c66:Write register start address

0x04:Write to 4 registers

0x08:Write 8 bytes

0x695064f9:timestamp 0x64f96950(1694067024)2023-09-07 14:10:24

0x851f40db: calibration value 0x40db851f (6.86)

0xf65c:Check digits

Received: 32 10 9c 66 00 04 0a 46

0x32:Device address

0x10:Write Multiple Register Function Code

0x9c66:Write register start address

0x04:Write to 4 registers

0x0a46:Check digit

3.4.4 pH3 calibration

Send: 32 10 9c 6a 00 04 08 6a b4 64 f9 e1 48 41 12 6d 71
0x32:Device address
0x10:Write Multiple Register Function Code
0x9c6a:Write register start address
0x04:Write to 4 registers
0x08:Write 8 bytes
0x6ab464f9:timestamp 0x64f96ab4(1694067380)2023-09-07 14:16:20
0xe1484112: calibration value 0x4112e148 (9.18)
0x6d71:check digit

Received: 32 10 9c 6a 00 04 ca 45
0x32:Device address
0x10:Write Multiple Register Function Code
0x9c6a:Write register start address
0x04:Write to 4 registers
0xca45:check digit

3.4.5 Ammonia 1 calibration

Send: 32 10 9c 72 00 04 08 a8 0a 64 f9 00 00 3f 80 88 f4
0x32:Device address
0x10:Write Multiple Register Function Code
0x9c72:Write register start address
0x04:Write 4 registers 0x08:Write 8 bytes
0xa80a64f9:timestamp 0x64f9a80a(1694083082)2023-09-07 18:38:02
0x00003f80:Calibration value 0x3f800000(1)
0x88f4:check digit

Received: 32 10 9c 72 00 04 4a 42
0x32:Device address
0x10:Write Multiple Register Function Code
0x9c72:Write register start address
0x04:Write to 4 registers
0x4a42:check digit

3.4.6 Ammonia 2 calibration

Send: 32 10 9c 76 00 04 08 a9 1e 64 f9 00 00 41 20 cd 2e
0x32:Device address
0x10:Write Multiple Register Function Code
0x9c76:Write register start address
0x04:Write to 4 registers
0x08:Write 8 bytes
0xa91e64f9:timestamp 0x64f9a91e(1694083358)2023-09-07 18:42:38
0x00004120:Calibration value 0x41200000(10)
0xcd2e:check digit

Received: 32 10 9c 76 00 04 0b 83
0x32:Device address
0x10:Write Multiple Register Function Code
0x9c76:Write register start address
0x04:Write to 4 registers
0x0b83:check digit

3.4.7 Temperature calibration

Send: 32 10 9c 7a 00 02 04 00 00 41 a0 4d 9e
0x32:Device address
0x10:Write Multiple Register Function Code
0x9c7a:Write register start address
0x02:Write to 2 registers
0x04:Write 4 bytes
0x000041a0:Thermometer measurement 0x41a00000(20)
0x4d9e:check digit

Received: 32 10 9c 7a 00 02 4b 82
0x32:Device address
0x10:Write Multiple Register Function Code
0x9c7a:Write register start address
0x02:Write to 2 registers
0x4b82:check digit

4 Use and maintenance

4.1 Probe activation

Before the probe can be used, it first needs to be activated.

Gently shake the sensor site as described in 3.1 to allow the electrolyte inside the electrode to refill the membrane head due to transport. The product should be kept upright for subsequent use.

Immerse the probe's electrode site in a 10-1000mg/L ammonia standard solution, keeping the temperature of the solution at the lower end of the temperature range of the water body to be measured, for 6 hours or more.

4.2 Calibration

4.2.1 pH calibration

The pH electrode is connected to the controller or host computer and the following steps are followed to complete the three-point calibration.

Nominal 4 pH standard solution point calibration: put the electrode into the nominal 4 pH buffer solution, wait until the test value is completely stable (generally 5-10min), in the "pH calibration value 1" at the current standard solution pH concentration (against the current temperature to get the standard value), and then in the "pH calibration time 1" to modify the corresponding time stamp for calibration, or in the controller at the "pH: calibration point 1" enter the current standard solution pH concentration (against the current temperature to get the standard value). calibration time 1", then modify the corresponding time stamp for calibration, or in the controller "pH: calibration point 1", enter the current pH concentration of the standard solution (against the current temperature to get the standard value), and then click on "calibration", wait for about 10s to complete the calibration, then click on "calibration". Wait for about 10s to complete the calibration of the nominal 4 pH standard solution point.

Nominal 7 pH standard solution point calibration: put the electrode into the nominal 7 pH buffer solution, wait until the test value is completely stable (generally 5-10min), in the "pH calibration value 2" at the current standard solution pH concentration (compared with the current temperature to get the standard value), and then modify the corresponding timestamp at the "pH calibration time 2" for calibration, or in the controller "pH: calibration point 2" at the current standard solution pH concentration (compared with the current temperature to get the standard value). calibration time 2", then in the "pH calibration time 2" to modify the corresponding time stamp for calibration, or in the controller "pH: calibration point 2", enter the current standard solution pH concentration (against the current temperature to get the standard value), and then click on "calibration", wait for about 10s to complete the calibration, then click on "calibration". Wait for about 10s to complete the calibration of the nominal 7 pH standard solution point.

Nominal 9 or 10 pH standard solution point calibration: put the electrode into the nominal 9 or 10 pH buffer solution, wait until the test value is completely stable (generally 5-10min), in the "pH calibration value 3" at the current standard solution pH concentration (against the current temperature to get the standard value), and then in the "pH calibration time 3" to modify the corresponding time stamp for calibration, or in the controller "pH: calibration point 3" at the current standard solution pH concentration (against the current temperature to get the standard value). calibration time 3", then in the "pH calibration time 3" to modify the corresponding time stamp for calibration, or in the controller "pH: calibration point 3", enter the current standard solution pH concentration (against the current temperature to get the standard value), and then click on "calibration", wait for about 10s to complete the calibration, then click on "calibration". Wait for about 10s to complete the calibration of the nominal 9 or 10 pH standard solution point.

4.2.2 Ammonia calibration

Ammonia calibration: according to the range of the water body to be measured to select the concentration of ammonia nitrogen to be measured range of 10% and 100% (or 80%) of the standard solution; for standards preparation, please use drinking water having no ammonia inside.

To maintain the temperature difference between the two standard solution is less than 0.5 °C, the sensor electrode partially submerged in the standard solution, to be stable after the number of (generally 5min to reach a stable) to enter the current standard ammonia nitrogen concentration for calibration; the same operation to complete the other point of the calibration.

Example:

The concentration of the test water body is 5.6mg/L, you can choose 1mg/L ammonium (NH_4^+) standard solution for point 1 and 10mg/L ammonium (NH_4^+) standard solution for calibration of point 2.

Gently take off the pH protective solution, then soak the sensor electrode partially in the 10 mg/L ammonium standard solution, leaving it for 5-10 min until the read measurement value would be stable.

Wash the electrode with pure water and gently dry it and then soak the sensor electrode partially in the 1 mg/L ammonium standard solution, then enter the concentration of 1 mg/L at the calibration point 1 after waiting for 5-10min.

Wash the electrode with pure water and gently dry it, then replace 10 mg/L ammonium standard solution and enter the concentration of 10 mg/L at the calibration point 2 after waiting for 5-10min.

Wash the electrode with pure water and gently dry it, then you can measure a $\text{NH}_3\text{-N}$ standard sample (i.e. 5 mg/L) prepared in tap water having very low NH_3 content to check the measured value.

4.3 Conditioning

After calibration, the sensor can measure, store and upload data normally. After replacing the liquid to be measured, it is necessary to re-stabilize the sensor for more than 5 minutes to ensure that the display is stable and then read the data.

4.4 Maintenance

Maintenance of the ion-selective electrode ammonia probe mainly includes cleaning, calibration, replacement of the ammonium ion electrode electrolyte and membrane cap.

Cleaning: In order to prevent impurities and microorganisms from adhering to the electrode surface and causing inaccurate measurements, it is necessary to clean the product once a week. Remove the sensor product from the water, use a clean non-woven cloth and distilled water to remove the ammonium ion electrode, temperature sensor and pH electrode surface dirt, be careful not to damage the electrodes with excessive force.

Calibration: Usually when the sensor is first used or used again after a long period of disuse, it is recommended to carry out a calibration at intervals of 1-2 days in the first 3 days of use after activation; it is recommended to carry out a calibration once a week after the subsequent electrode has been stabilized in use.

Maintenance of Ammonium Ion Electrode: In order to keep the better performance of the ammonium ion electrode, it is recommended to replace the electrolyte once every 3-6 months and the membrane cap once every 6-10 months. Replacement methods are as follows:

- Unscrew the electrode cap of the ammonium ion electrode;
- pour out the electrolyte remaining in the electrode body;
- use a pipette to place fresh electrolyte into the electrode body until the electrolyte forms a bump (capillary action).
- Wipe up any spilled electrolyte;
- Screw the electrode cap back onto the electrode body;
- Slowly rotate the electrode with the head facing down. This process can remove the air bubbles in the head; then slightly force the electrode vertically downward flinging process to ensure that the electrode head down,

and not too hard, otherwise it will cause damage to the electrode cap.

- Always keep the electrode upright and facing downwards before installing it back into the holder, thus avoiding air bubbles back at the electrode cap.

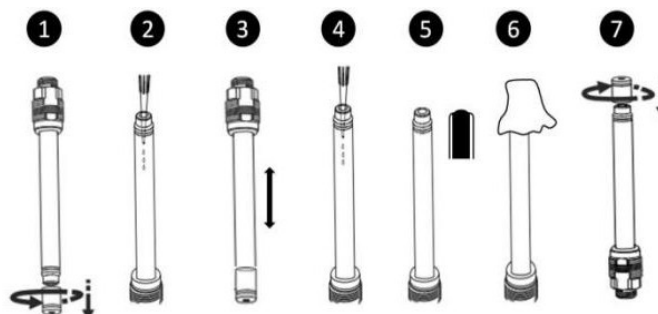


Figure 4.1 Ammonium ion electrode replacement membrane cap and electrolyte operation

4.5 Troubleshooting

If the instrument is not installed, wired correctly according to the user's manual, improper use or component damage may cause instrument failure, Table 2 gives the instrument a few

Typical faults and troubleshooting methods. When there is a problem with the instrument, please check which kind of trouble belongs to in Table 2 first, and then solve it one by one according to the tips; if the problem is complicated, please contact us for support in time, and do not try to solve the problem by yourself.

Table 4.1 Typical faults and handling

serial number	fault phenomenon	Possible causes	exclusion method
1	Ammonia concentration is constant at 0 or abnormally high; temperature and pH readings are normal.	Failure of ammonium ion electrode	Firstly, observe whether there is any damage to the ammonium ion electrode body and the membrane head, flick the sensor electrode tip to remove the air bubbles at the membrane head; refill the ammonium ion electrode electrolyte and replace the membrane cap; Disassembly to check circuit connections and boards
		Bubbles at the ammonium ion electrode membrane	
		Insufficient electrolyte	
		Internal circuit connection failure	
		Circuit board damage	
2	Ammonia nitrogen concentration readings via even after full stabilization However, there are big fluctuations.	Large external electromagnetic interference	Firstly, determine whether there is a large disturbance in the water body and, if necessary, take the The sensor is placed in a stationary standard solution for testing and observation; and then
		Measurement of high water disturbance	
		Aging of ammonium ion electrode	
3	Abnormal pH reading	pH Electrode ageing pH electrode damage circuit failure	Firstly, observe the pH electrode body and the head of the membrane for breakage. Damage, whether the internal blackening; disassembly to check the circuit connections and circuit boards

6 Additional information

6.1 Unpacking and inspection

When you receive the purchased instrument to open the box before the inspection, please carefully check the packaging for any damage or signs of falling, if there is any suspicion and cause damage to the contents of the goods, quickly notify the company and the carrier! Please be careful when opening the box, do not damage the instrument or other parts. After opening the box, please check each piece, if you find the wrong shipment, missing or damaged, please notify the company as soon as possible.

The pH sensor is shipped covered by a tap including a KCl 3M buffer solution; in order to use it on site, please remove the tap.

6.2 Safekeeping

When the instrument you purchased is not to be installed urgently or needs to be put aside for a longer period of time, please store it properly and do not put it in the following places:

- Ambient temperature below 5°C or above 40°C;
- Direct sunlight;
- Next to the heater;
- The environment is vibrating and dusty;
- Keep the pH electrode in a moist protective sleeve as long as possible prior to installation.

When you have purchased an instrument that has been installed but will not be used for an extended period of time, please clean the electrode with DI water, keep the pH probe protective sleeve in place, remove the probe sleeve when activated and close the pH sensor with the same shipment tap filled up by KCl 3 M buffer.

6.3 Precautions to use

★ Avoid rapid insertion into a place that is not connected to the outside atmosphere, which may cause air bubbles to enter the head of the film.

★ Use to avoid prolonged exposure of the probe to dry environments caused by water cuts.

6.4 After sale service

When during the installation, commissioning and use of the process would be encountering problems, please consult our company in a timely manner, especially found fault, please as far as possible will be the instrument number, model number, date of purchase, failure phenomenon described clearly so that we can quickly and accurately to provide you with support.